MSG No. 121.328 $I\bar{4}2m1'$ [Type II, tetragonal]

Table 1: Wyckoff site: `2a`, site symmetry: `-42m1'`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,2,3,4,5,6,7,8,17,18,19,20,21,22,23,24]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[9,10,11,12,13,14,15,16,25,26,27,28,29,30,31,32]`

Table 2: Wyckoff site: `2b`, site symmetry: `-42m1'`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[1,2,3,4,5,6,7,8,17,18,19,20,21,22,23,24]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[9,10,11,12,13,14,15,16,25,26,27,28,29,30,31,32]`

Table 3: Wyckoff site: `4c`, site symmetry: `222.1'`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ 0\right]$	`[1,2,3,4,17,18,19,20]`
2	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[5,6,7,8,21,22,23,24]`
3	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[9,10,11,12,25,26,27,28]`
4	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[13,14,15,16,29,30,31,32]`

Table 4: Wyckoff site: `4d`, site symmetry: `-4..1'`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[1,4,13,14,17,20,29,30]`
2	$\left[0,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[2,3,15,16,18,19,31,32]`
3	$\left[\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	`[5,6,9,12,21,22,25,28]`
4	$\left[\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	`[7,8,10,11,23,24,26,27]`

Table 5: Wyckoff site: `4e`, site symmetry: `2.mm1'`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,4,7,8,17,20,23,24]`
2	$[0,\ 0,\ -z]$	`[2,3,5,6,18,19,21,22]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[9,12,15,16,25,28,31,32]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[10,11,13,14,26,27,29,30]`

Table 6: Wyckoff site: `8f`, site symmetry: `.2.1'`

No.	position	mapping
1	$[x,\ 0,\ 0]$	`[1,2,17,18]`
2	$[-x,\ 0,\ 0]$	`[3,4,19,20]`
3	$[0,\ -x,\ 0]$	`[5,7,21,23]`

continued ...

Table 6

No.	position	mapping
4	$[0,\ x,\ 0]$	`[6,8,22,24]`
5	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[9,10,25,26]`
6	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[11,12,27,28]`
7	$\left[\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[13,15,29,31]`
8	$\left[\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[14,16,30,32]`

Table 7: Wyckoff site: `8g`, site symmetry: `.2.1'`

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{2}\right]$	`[1,2,17,18]`
2	$\left[-x,\ 0,\ \frac{1}{2}\right]$	`[3,4,19,20]`
3	$\left[0,\ -x,\ \frac{1}{2}\right]$	`[5,7,21,23]`
4	$\left[0,\ x,\ \frac{1}{2}\right]$	`[6,8,22,24]`
5	$\left[x+\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[9,10,25,26]`
6	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ 0\right]$	`[11,12,27,28]`
7	$\left[\frac{1}{2},\ \frac{1}{2}-x,\ 0\right]$	`[13,15,29,31]`
8	$\left[\frac{1}{2},\ x+\frac{1}{2},\ 0\right]$	`[14,16,30,32]`

Table 8: Wyckoff site: `8h`, site symmetry: `2..1'`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	`[1,4,17,20]`
2	$\left[0,\ \frac{1}{2},\ -z\right]$	`[2,3,18,19]`
3	$\left[\frac{1}{2},\ 0,\ -z\right]$	`[5,6,21,22]`
4	$\left[\frac{1}{2},\ 0,\ z\right]$	`[7,8,23,24]`
5	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	`[9,12,25,28]`
6	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-z\right]$	`[10,11,26,27]`
7	$\left[0,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[13,14,29,30]`
8	$\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[15,16,31,32]`

Table 9: Wyckoff site: `8i`, site symmetry: `..m1'`

No.	position	mapping
1	$[x,\ x,\ z]$	`[1,8,17,24]`
2	$[x,\ -x,\ -z]$	`[2,5,18,21]`
3	$[-x,\ x,\ -z]$	`[3,6,19,22]`
4	$[-x,\ -x,\ z]$	`[4,7,20,23]`
5	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[9,16,25,32]`
6	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	`[10,13,26,29]`
7	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	`[11,14,27,30]`
8	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	`[12,15,28,31]`

Table 10: Wyckoff site: `16j`, site symmetry: `11'`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1,17]`
2	$[x,\ -y,\ -z]$	`[2,18]`
3	$[-x,\ y,\ -z]$	`[3,19]`
4	$[-x,\ -y,\ z]$	`[4,20]`
5	$[y,\ -x,\ -z]$	`[5,21]`
6	$[-y,\ x,\ -z]$	`[6,22]`
7	$[-y,\ -x,\ z]$	`[7,23]`
8	$[y,\ x,\ z]$	`[8,24]`
9	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[9,25]`
10	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	`[10,26]`
11	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	`[11,27]`
12	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	`[12,28]`
13	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	`[13,29]`
14	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	`[14,30]`
15	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	`[15,31]`
16	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[16,32]`